

Figure 17.12 Ascidian

Subphylum Cephalochordata

Cephalochordates are small fish like marine chordates with unpaired dorsal fins. The notochord extends throughout the entire length of the body. E.g. *Amphioxus*

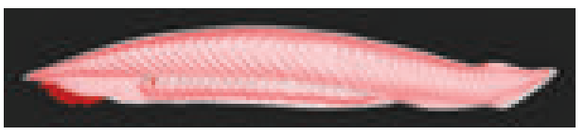


Figure 17.13 Amphioxus

17.3.2 Vertebrata

This group is characterized by the presence of vertebral column or backbone. Notochord in an embryonic stage gets replaced by the vertebral column, which forms the chief skeletal axis of the body. Vertebrata are grouped into six classes.

Class: Cyclostomata

Cyclostomes are jawless vertebrates (mouth not bounded by jaws). Body is elongated and eel like. They have circular mouth. Skin is slimy and scaleless. They are ectoparasites of fishes. E.g. *Hagfish*.



Figure 17.14 Lamprey

Class: Pisces

Fishes are poikilothermic (**cold-blooded**), aquatic vertebrates with jaws. The **streamlined body** is divisible into head, trunk and tail. Locomotion is by **paired** and **median fins**. Their body is covered with **scales**. Respiration is through **gills**. The heart is two chambered with an auricle and a ventricle. There are two main types of fishes.

(i) **Cartilaginous fishes**, with skeleton made of cartilages e.g. Sharks, Skates.

(ii) **Bony fishes** with skeleton made of bones e.g. Carps, Mullets.



Figure 17.15 Shark

More to Know



The smallest vertebrate, Philippine goby/dwarf pygmy goby is a tropical species fish found in brackish water and mangrove areas in south East Asia, measuring only 10 mm in length.

Class: Amphibia (amphi- both; bios- life)

These are the first four legged (**tetrapods**) vertebrates with **dual adaptation** to live in both land and water. The body is divisible into head and trunk. Their skin is moist and have **mucus glands**. Respiration is through **gills, lungs, skin** or **buccopharynx**. The heart is three chambered with two auricles and one ventricle. Eggs are laid in water. The **tadpole larva**, transforms into an adult. e.g-Frog, Toad.





The Chinese giant salamander *Andrias davidians* is the largest amphibian in the world. Its length is about five feet and eleven inches. It weighs about 65 kg, found in Central and South China.

IX_SCI_EM_Unit-17_BIO.indd 205 06-02-2020 13:06:11